

Daily Geometry Drill #2

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Circle Equations (centre $(-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$) · Tangents · Line–Circle Tests · Chord Lengths.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Complete the square, read the centre and radius, then write or test circle data on sight.

- Write down the centre of the circle $x^2 + y^2 - 6x - 8y + 24 = 0$.
- Write down the radius of the circle $x^2 + y^2 - 6x - 8y + 24 = 0$.
- Write down the equation of the circle with centre $(2, -1)$ and radius 5.
- Write down the radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$.
- The point $(x, 4)$ lies on the circle $x^2 + y^2 = 25$ with $x > 0$. Find x .
- The radius from the origin to the point $(3, 4)$ on $x^2 + y^2 = 25$ has what gradient?
- The tangent to $x^2 + y^2 = 25$ at the point $(3, 4)$ has what gradient?
- Write down the equation of the circle with centre $(-1, 2)$ and radius 3.
- The centre of $x^2 + y^2 - 6x - 8y + 24 = 0$ is $(3, 4)$. Find the perpendicular distance of this centre from the line $4x + 3y - 9 = 0$.
- A chord of the circle $x^2 + y^2 = 25$ is at perpendicular distance 3 from the origin. Find the length of the chord.

Section B Manipulation Drills

(≤50 s each)

Combine the new circle tools with last drill's line machinery. Complete the square first, then test.

- Which of the following is the equation of the circle with centre $(3, -2)$ and radius 4?
 - $x^2 + y^2 - 6x + 4y - 3 = 0$
 - $x^2 + y^2 - 6x + 4y + 9 = 0$
 - $x^2 + y^2 + 6x - 4y - 3 = 0$

- D) $x^2 + y^2 - 6x - 4y - 3 = 0$
2. The circle $x^2 + y^2 + 2x - 4y - 11 = 0$ has what radius?
- A) 2
B) 3
C) 4
D) $\sqrt{5}$
3. The point $(1, 2)$ lies on the circle $x^2 + y^2 = 5$. The equation of the tangent to the circle at $(1, 2)$ is?
- A) $y = -\frac{1}{2}x + \frac{5}{2}$
B) $y = 2x$
C) $y = -2x + 5$
D) $y = \frac{1}{2}x$
4. A circle of form $x^2 + y^2 + 2gx + 2fy + c = 0$ passes through the origin. Deduce c .
- A) $c = 1$
B) $c = 0$
C) $c = g + f$
D) $c = -1$
5. The perpendicular distance from the centre of $x^2 + y^2 = 25$ to the line $y = 2x + 3$ is?
- A) $\frac{3}{\sqrt{5}}$
B) $\frac{3}{5}$
C) 3
D) 2
6. Which of the following is a tangent to the circle $x^2 + y^2 = 4$?
- A) $y = x + 2\sqrt{2}$
B) $y = x + 4$
C) $y = 2x + 2$
D) $y = 2x + 4$
7. A chord of the circle $x^2 + y^2 = 13$ has length $4\sqrt{3}$. What is the perpendicular distance of the chord from the centre?
- A) 3
B) 2

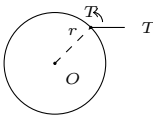
- C) $\sqrt{5}$
D) 1
8. The line $y = mx + 4$ is a tangent to the circle $x^2 + y^2 = 4$. Find the two possible values of m .
A) ± 1
B) $\pm\sqrt{3}$
C) ± 2
D) $\pm\sqrt{2}$
9. The circle with centre $(6, a)$ and radius 6 is tangent to the y -axis and passes through the point $(0, 8)$. Find a .
A) 1
B) 3
C) 5
D) 8
10. Write down the length of the chord cut from the line $y = x + 1$ by the circle $x^2 + y^2 = 25$.

Section C Substitution & Structure(≤ 90 s each)

Hard: hidden tangency, non-obvious chord bisection, and the distance-vs-discriminant choice. No brute substitution.

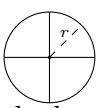
1. (after TMUA 2017 Paper 1 Q6) A tangent to the circle $x^2 + y^2 = 144$ passes through $(20, 0)$ and meets the positive y -axis at T . What is the y -coordinate of T ?
A) 12
B) 15
C) $\frac{49}{3}$
D) 20
2. (adapted from TMUA 2018 Paper 1 Q7) The point $P(8, 6)$ is outside the circle $x^2 + y^2 = 36$. Two tangents from P touch the circle at A and B . What is the length AB (the common chord of contact)?
A) $\frac{12\sqrt{13}}{13}$
B) $\frac{24\sqrt{13}}{13}$
C) $\frac{48}{5}$
D) $\frac{72}{5}$
3. (after TMUA 2019 Paper 1 Q6) The circles $(x + 4)^2 + (y + 1)^2 = 64$ and $(x - 8)^2 + (y - 4)^2 = r^2$ have exactly one point in common. If $r < 13$, what is r ?
A) 5

- B) 8
C) 13
D) 21

4.  (adapted from TMUA 2017 Paper 1 Q9) A circle has equation $x^2 + y^2 - 18x - 22y + 178 = 0$ (centre $(9, 11)$, $r = 5$). A regular hexagon $ABCDEF$ is inscribed so that all vertices lie on the circle. What is the area of the hexagon?

- A) $18\sqrt{3}$
B) $36\sqrt{3}$
C) $\frac{75\sqrt{3}}{2}$
D) 75

5. The line $y = 2x + 5$ is a tangent to the circle $x^2 + y^2 = r^2$. Which r^2 makes the distance from the centre to the line equal to r ?
- A) 1
B) 5
C) 20
D) 25
6. (after MAT 2019 Q6) The circle $x^2 + y^2 - 4x + 2y - 11 = 0$ and the line $x - y + 1 = 0$: determine the number of intersection points without solving the system.
- A) 0 (line misses)
B) 1 (tangent)
C) 2 (secant)
D) Infinitely many (line is diameter)

7.  The circle $x^2 + y^2 - 8x - 4y - 21 = 0$ ($r = \sqrt{41}$) is cut by the y -axis ($x = 0$). What is the length of the chord intercepted on the y -axis?
- A) $2\sqrt{21}$
B) $2\sqrt{41}$
C) 8
D) 10

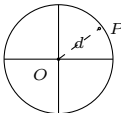
8. (after TMUA 2020 Paper 1 Q7) The line $y = kx + 5$ is tangent to the circle $x^2 + y^2 - 6x - 8y + 24 = 0$ (centre $(3, 4)$, $r = 1$). How many distinct k satisfy this?
- A) 0

- B) 1
- C) 2
- D) Infinitely many

Section D Challenge

(TMUA / SMC / BMO1 difficulty)

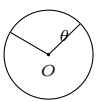
Hardest 5: distance invariants, hidden right angles, and a construction you must spot.

1.  (adapted from TMUA 2018 Paper 1 Q7 / 2022 Paper 1 Q3) The circles $C_1 : (x + 2)^2 + (y - 3)^2 = 18$ and $C_2 : (x - 7)^2 + (y + 6)^2 = 2$ have centres O_1, O_2 . What is the *shortest* distance between a point on C_1 and a point on C_2 ?

- A) $5\sqrt{2} - 4$
- B) $5\sqrt{2} - 5$
- C) $5\sqrt{2}$
- D) $5\sqrt{2} + 5$

2. (after TMUA Specimen Paper 1 Q9) The segment joining $(3, 3)$ and $(7, 5)$ is a diameter of circle C . C is translated 3 left, reflected in the x -axis, then enlarged scale factor 4 about its centre. What is the equation of the final circle?

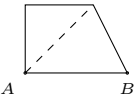
- A) $(x - 2)^2 + (y - 4)^2 = 80$
- B) $(x - 2)^2 + (y + 4)^2 = 80$
- C) $(x - 2)^2 + (y - 4)^2 = 320$
- D) $(x - 2)^2 + (y + 4)^2 = 320$

3.  (adapted from TMUA 2022 Paper 1 Q14) Circle O radius 6, $\angle POQ \geq \pi/2$, area $[POQ] = 9\sqrt{3}$. For variable R on the major arc PQ , what is the *greatest possible* area of $\triangle PRQ$?

- A) $18 + 9\sqrt{3}$
- B) $27\sqrt{3}$
- C) $27 + 9\sqrt{3}$
- D) $36 + 9\sqrt{3}$

4. (after MAT 2020 Q1H) Let $P(p, q)$ and circle $C : x^2 + 2fx + y^2 + 2gy + h = 0$ with centre $(-f, -g)$ and $L = |PC_0|$. Which set is *minimal sufficient* to compute L ?

- A) f, g, h
- B) f, g, p, q
- C) f, h, p, q
- D) g, h, p, q

5.  (adapted from BMO1 2010 Q4) $ABCD$ is a cyclic quadrilateral with perpendicular diagonals intersecting at E . The line through E perpendicular to AB meets CD at F . Which statement is *necessarily true*?

- A) F is the midpoint of CD (Brahmagupta)
- B) F bisects $\angle CED$
- C) $EF = AB/2$
- D) $ABCD$ must be an isosceles trapezium

“Tangency is a distance statement in disguise: compare d to r , never solve the intersection.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths